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BT102-Microbiology

Current paper solved question final term February 2020

1. Define and type of algae. 5 marks

Define :

The definition of algae is a single or multi-cellular organism that has no roots, stems or leaves and is often found in water

These are simple eukaryotic cells. Some are unicellular, others are multicellular (thallus); however, they lack tissues such as roots, stem and leaves typically seen in plants. Algae absorb nutrients from water through their surfaces and are mostly photoautotrophs; however, a few are chemoheterotrophs. They are responsible for 80% atmospheric O2 on the face of the earth.

Types of algae:

1. **Phaeophyta (Kelp):** Commonly known as brown algae, brownish in color, contains cellulose and alginic acid cell walls, are multicellular and contain chlorophyll *a* and *c*, xanthophylls. These store carbohydrates and are used for algin which is a thickener in ice creams. Can grow upto 20 cm in one day.

2. **Rhodophyta:** Red Alage, these are reddish in color and contain cellulose in their cell walls. Most are multicellular and contain chlorophyll a and d, and phycobiliproteins. They store glucose polymer and used for agar and carrageenan (both are used as thickeners of foods).

3. **Chlorophyta:** Green Algae: have all the features of red algae except that chlorophyll is of a and b types only. These algae are believed to have given rise to terrestrial plants.

4. **Bacillariophyta:** Cell wall is composed of pectin and silica. They are unicellular and possess chlorophyll a and c, and carotene xanthophylls. These algae store oil and produce domoic acid with which human can be intoxicated by eating contaminated mussels. **Diatoms**: Come in beautiful colors and shapes.

5. **Dinoflagellata:** These are unicellular algae collectively called as plankton (free floating organisms). These cells have cellulose in their cytoplasmic membrane which gives the membrane rigidity. They have chlorophyll *a* and *c*, and carotene. These algae store starch in them. Dinoflagellates in the genus Alexandrium produce neurotoxins (called saxitoxins) which can cause paralytic shellfish poisoning in humans when they consume contaminated mussels and clams. Blooms of Alexndrium gives ocean a deep red color commonly known as red tide.

6. **Oomycota (Water molds):** They are basically decomposers of dead matter. Superficially, these resemble with fungi; however, these are motile with flagella and their cell wall is composed of cellulose not chitin. Their spores are called zoospores. Terrestrial members of water molds are also plant parasite. *Phytophthora infestans* was responsible for Irish potato blight during mid-1800s. *P. ramorum* causes sudden oak death

2. HYPERSENSITIVITY AND ITS TYPE 5 marks

What is hypersensitivity?

It is a normal Immune response but in a damaging way. In other words, immune system is acting in a way for which it has made to. So when an antigen gets into the body, immune system recognizes it as foreign and start making antibodies against the antigen. However, the consequences are bad and damaging to the body. It may be called an abnormal antigenic response. Allergy is another name for hypersensitivity. Those antigens that act in a bad way are called allergens

Types of hypersensitivity:

Hypersensitivities can be grouped into 4 types based on the types of antibodies or cells involved.

1. **Type I (Anaphylactic):** Also called anaphylaxis or immediate hypersensitivity. This is IgE mediated.
2. **Type II (Cytotoxic):** IgG and IgM mediated as these antibodies attach to cells and destroy them by activating complement.
3. **Type III (Immune complex mediated):** Immune complexes are formed by IgM or IgG antibodies and these complexes lodge in the capillary beds and cause damage by complement activation.
4. **Type IV (delayed type cell-mediated):** CD4+ helper cells along with macrophages are involved.

3. Define antigen and epitope

1. **Antigen**: A foreign substance that provokes immunity is called an antigen.
   The definition of antigen is a harmful substance which enters the body which causes the body to make antibodies as a response to fight off disease.
   More specifically called as an immunogen. So immunogen and antigen although slightly different but used interchangeably. o Proteins are best antigen, though glycoproteins and lipoproteins can also act as antigens. Carbohydrates and lipid alone are poor antigens. o Antigens are foreign to the body. o **Degradability**: Inert molecules are not antigenic.
2. **Epitopes**: Specific regions on an antigen. In other words, any antigen has multiple regions each one of those act as a small antigen. These are called epitopes.These epitopes are also called as antigenic determinants. o Most antigens have MW >10,000 D. Molecules that are less than 10,000 D are generally poor antigen.

   A small molecular region of an antigen that binds to a particular antibody or antigen receptor on a T cell; an antigenic determinant. A single antigen can have multiple epitopes

4. Describe two transported system in the blood and why it is necessary 10 marks

Cardiovascular and **lymphatics** are the two transport systems of the body by which cells of the immune system patrol the whole body.

Cardiovascular system consists of the heart and associated blood vessels (arteries and veins) in which blood circulates

lymphatics are channels that start from tissue spaces and collect tissue fluid (called lymph) from tissue spaces and bring them back to the heart. Along these lymphatic channels, there are strategically placed lymph nodes which filter lymph for microbes much like blood is filtered in the spleen. Lymph nodes contain aggregations of B and T cells along with macrophages and dendritic cells which interact with each other inducing adaptive immune responses.

Why Cardiovascular and **lymphatics are needed?**

Well, in unicellular organisms, nutrients can be absorbed directly from the environment in which a unicellular organism is living. However, a multicellular organism does not have this easy access to the nutrients. So, nutrients must surround the cells in a multicellular organism. These nutrients are provided by cardiovascular system which pumps blood to all organs. Remember that only fluid part of blood leaves blood vessels. RBCs never leave blood vessels. Sometimes proteins also leak from these blood vessels; however, once they leak from the blood vessels, they never can enter back into blood vessels because of their bigger size. If proteins remain in the tissue spaces, they exert osmotic pressure which attracts water into these space resulting in edema. However, nature has provided another mechanism in the form of lymphatic channels which start from tissue spaces and can drain tissue fluid and proteins back to the heart. This is the role of blood vessels and lympahtics that they ensure

circulation of nutrients throughout the body without causing any edema or abnormality. B cells and T cells also keep circulating between these two channels (cardiovascular and lymphatics). Unfortunately, microbes can also use these two systems to spread themselves in the body. That is the reason, spleen and lymph nodes are placed in these channels by the nature to deal with these microbes that find their ways into blood and lymph. ? Lymphatic System consists of the following: o Bone marrow o Lymph nodes o Spleen o Tonsils o Thymus o Mucosa associated o Bursa of Fabricius in birds

5. Disk diffusion method

A standardized inoculum of the organisms is spread on agar plate. Filter paper disks soaked with anti bacterials are then placed on the lawn of the organism. Plates are incubated at 37oC and results are recorded. Drugs diffuse from the paper disks radially and if the organisms are sensitive to the drug, a clear area with no growth around the disks develops. The size of the zone directly relates to the degree of sensitivity of the drug. This zone is compared to a standard table for recording the results of sensitivity test. Results are reported as sensitive or resistant or intermediate (between the two extremes).

6. How microbes develop resistance in antimicrobial?

7. Define immunity and difference between innate immunity and adaptive immunity. 10 marks

What is immunity?

Immunity: Ability of the body to protect itself from harmful effects of diseases.

Susceptibility: lack of immunity which leads to easy establishment of disease.

Types of immunity

Innate (Natural):

Present at birth: One is born with these defenses, so it is present without any exposure to the microbes. That is the reason, it is non-specific.

Rapid to respond: Because it is already there, it interacts with the microbes as soon as they enter the body.

Nonspecific: It means that it does not differentiate if it is an E. coli or a Staph or Salmonella. Body defense system against these entire pathogens act in a similar fashion. In other words, the same macrophage can phagocytize E. coli, Staph or Salmonella at the same time (if these pathogens are present there).

Does not increase in response with repeated exposures to the same agent. Therefore, it has no memory.

First line of defense: Physical barriers (skin, mucus membrane) provide the first line of defense as they prevent the entry of the organisms by providing a physical barrier which if broken will let the microbes go inside.

Second line of defense: Phagocytic cells and their secretions provide a second line of barriers for microbes. So, if microbes succeed in breaching or breaking the first barrier, macrophages, neutrophils and such other cells phagocytize these microbes and kill them before these can establish infection in the body.

Adaptive Immunity: This will be studied in detail in a separate lecture soon.

Adaptive (Acquired): It is the third line of defense. Acquired by experience, specific to an organism.

Specific: It means that antibodies made against E. coli will destroy E. coli only, and will not do any harm to a Staph for example.

Have memory: It means if antibodies have once been made against E. coli, this exposure will be remembered by the body, and if a second encounter is made with the same type of E. coli again, the body would mount a heightened immune response against E. coli because it has memory cells developed in the body which upon second stimulation proliferate quickly and mount a greater defensive response.

O Because of memory, body's immune response increase with every repeated exposure. It has a limit also!

Slower to respond: For the first time exposure, the body's response is slower and takes a long time compared with the innate immunity which is immediate.

Adaptive Immune Response can further be divided into:

Humoral immune response (antibody response by B cells or lymphocytes)

Cellular immune response (T cells are made which kills microbial harboring cells).

8. Agglutination and Neutralization 2 marks

Agglutination: Antibodies can agglutinate antigens which are then cleared as a group by macrophages

Neutralization: Toxins bind with antibodies and they become incapable of binding to their target receptors thus the toxins become harmless.

9. What is plaque 2 marks

Plaques are equivalent to bacterial colonies. Basically a plaque is an area where the virus infection has lysed the cells. These plaques are formed on cultured cells in a culture dish. A monolayer of the cells is infected with the virus to be enumerated and then the culture is overlaid with the semisolid agar so that infected and lysed cells do not move away from the site in the dish. Plaques can be counted easily much like bacterial colonies

10. Difference between capsule and slime

Capsule: If glycocalyx is organized and firmly attached to the cell, it is called a capsule.

– Plays important role in virulence (degree of pathogenicity)

– Can also be a part of the vaccine against the bacteria to which it belongs.

– Can be demonstrated by negative staining technique.

Slime: If glycocalyx is loosely attached to the cell, it is called slime.

11.PCR method 3 marks

PCR goes through 3 step cycles for about 30 to 35 times. Each cycle consists of a denaturing stage (about 95oC), primer annealing (50 to 56oC) and extending stage (72oC). A primer is a fragment of DNA which is specific to the signature sequence. Two primers span the signature sequence actually. At the end of 30 or so such cycles, enough DNA

accumulates that one can visualize on an agarose gel easily.

12. Type of T cell

T helper cells

TH1 (CD4+)

TH2 (CD4+)

T cytotoxic cells (CD8+)

13. Inhibits growth cell wall drug name

Penicillins and cephalosporins are the major antibiotics that **inhibit** bacterial **cell wall** synthesis. They are called beta-lactams because of the unusual 4-member ring that is common to all their members.

14. Microbiota factor and distribution composition

Nutrients: Availability of various nutrients determines the type of organisms present in a system. Similarly, physical and chemical factors also influence the type of microbiota. Other factors include pH, O2, and CO2. Host defenses, level of stress and age of an individual also matters for allowing microbiota of an individual.

15. Capsule function 2 marks

The capsule impairs phagocytosis and help the organisms evade the immune system; however, antibodies can help cells of the immune system in phagocytosis of such encapsulated organisms. Only a few bacteria produce capsule which include Strep pneumonia, Klbesiella pneumonia.

16. Viral replication stages

Viral multiplication goes through various stages in the cell. These include viral attachment, entry into the cell by pinocytosis or fusion with the plasma membrane, uncoating of the virus, biosynthesis of capsids and nucleic acids, maturation (assembly of the virus) and final release of the virus from the cell.

17. Economic benefits of fungi 3 marks

Saccharomyces are used for alcoholic beverages under anaerobic conditions for alcohol production (fermentatively). However, if Saccharomyces are incubated aerobically (for bread making), they metabolize glucose to produce CO2 (some ethanol is also produced but gets evaporated during baking) which expands the dough causing it to rise. This yeast is also extensively used for molecular biology work and vaccine production. Vitamin C is obtained from Aspergillus niger; cellulase from Trichoderma, antibiotics from many fungi and anticancer Taxol from Taxomyces are some other examples of useful fungal products. Entomophaga (kills gypsy moth), hence is used as a biological pest control. Mushrooms can be cooked and eaten as a source of proteins for humans.

18.Types of symbiosis 3 marks

Interaction of microbiota with the host leads to a relationship called symbiosis. There are a few outcomes of this symbiosis which we must know. The two participants of this relationship are microbiota and the host. At least one of these participants is benefited and sometimes both can be benefited from this mutual relationship. Different names or terms have been given to these mutual relationships as under:

Mutualism: both are benefited

Parasitism: one is harmed.

Commensalism: One is benefited, other is unharmed.

19. Inflammation and write its consequence 5 marks

Damage to body tissues triggers a local defensive response which is called inflammation. It causes redness, pain, swelling, heat, and loss of function of that organ

During inflammation, various mediators of inflammation are released, some of which include:

Histamine

Kinins: Almost all cells.

Prostaglandins by damaged cells

Leukotrienes by mast cells

Cytokines are released (TNF-alpha), and they cause release of acute phase proteins from the liver

20. Cause and diagnose of anaphylaxis

The patient also experiences difficulty in breathing. It is an emergency as it may kill the person. Anaphylaxis may be local or systemic.

o Shock from drug reactions, venoms, and common allergens causing asthma are some examples of type I hypersensitivity.

Hypersensitivity can be diagnosed by demonstration of IgE by intradermal injections of allergens one by one. If specific IgE are present in the serum, a reaction will be visible on the skin locally in the form of a wheel of inflammation. Moreover, IgE levels can be measured by ELISA. Measurement of IgE is more practical as nature of allergens is not known most of the time.

o Rx: Epinephrine & antihistaminic drugs provide a means of treatment.

21. Two name of bacteria which is capsulated

Bacillus anthracic, streptococcus pneumoniae and Klebsiella are examples of capsulated organisms.

22. What is Epidemiology and its types

The study of where and when diseases occur and how they are transmitted in a population.

There are three basic types of investigations relating to epidemiology

1. Descriptive Epidemiology
2. Analytical Epidemiology
3. Experimental Epidemiology:

23. Target site of antibiotics

24. Capsomere

Capsid is made up of protein subunits called capsomeres that are arranged in 3 possible ways to give a helical, or icosahedral or complex symmetry to the viruses.

 Capsomeres: Subunit called protomers aggregate to form capsomeres.

25. Character of antigen

Antigen: A foreign substance that provokes immunity is called an antigen.

More specifically called as an immunogen. So immunogen and antigen although slightly different but used interchangeably.

Proteins are best antigen, though glycoproteins and lipoproteins can also act as antigens. Carbohydrates and lipid alone are poor antigens.

Antigens are foreign to the body.

Degradability: Inert molecules are not antigenic.

26. Algae structure

Body of multicellular alga such as seaweed is called a thallus which consists of branched holdfasts (anchor alga to rock) stemlike hollow stipes and leaflike blades. There is no vascular tissue in these algae. Also, the stipe is not lignified or woody, so it does not provide support to the weed. Surrounding water provides the support for the thallus. Some algae have a gas filled body inside them which keeps them floating in the water. This gas filled structure is called pneumatocyst or float

27. Acid fast stanning 5 marks

Some bacteria such as Mycobacteria have a waxy material in their cell wall.

Specific name for this waxy substance is mycolic acid

Mycobacterium tuberculosis: Causative agent for tuberculosis

M. leprae: Causative agent for leprosy

• **Principle of Acid-fast staining**: Mycobacteria are lipophilic, not easy to stain

But once stained, they are resistant to acidic alcohol decolorization process.

• **Overall Procedure for Acid-fast staining**:

Carbol fuchsin: It is a primary stain Heat is used to enhance penetration of carbol fuchsin.

Acid alcohol is a decolorizer.

Methylene blue is a counterstain.

Please note, Mycobacteria are gram positive, however, they look red in acidfast staining.

28. Capsule staining 10 marks

Capsule: If glycocalyx is organized and firmly attached to the cell, it is called a capsule.

– Plays important role in virulence (degree of pathogenicity)

– Can also be a part of the vaccine against the bacteria to which it belongs.

– Can be demonstrated by negative staining technique.

– Capsulated organism make glistening colonies as seen in the accompanying diagram below:

 Bacillus anthracic, streptococcus pneumoniae and Klebsiella are examples of capsulated organisms.

 Thickness of the capsule depends upon the culture conditions.

Capsules are mostly are water soluble. Also remember that capsular organisms usually make the broth viscous and stringy

Capsule staining procedure

1. Prepare a thick smear in a loopful of congo red (1%) stain
2. Fixation in acid alcohol for 15 sec
3. Wash with dH2O
4. Cover it with acid fuchsin for 1 min.
5. Wash with water
6. A bacterium stains red, capsule remains colorless and is seen in the dark blue background as seen in the accompanying diagram.

29. Define vaccine and its types 10 marks

What is vaccine:

A product that creates immunity against an infection is called a vaccine. It could be the whole organism or a part of the organism.

The observation that people or animals once recover from an infection do not get sick easily to that organism again led to the discovery of vaccines and vaccination.

Types of vaccine:

Live Attenuated Vaccines: These are weakened organisms so that they do not harm the body, but divide or replicate in the same way as pathogenic version of the organism does in the body. So, when a live virus or bacteria multiplies in the body, it stimulates the immune system to create an adaptive immune response which is protective against the disease for which vaccine was given.

The benefits of live vaccines are a better immunity because live microbes mimic actual infection (although they do not cause harm to the body). Live vaccines induce both humoral and cellular immune responses together, so live vaccines

are a better choice if available. Because live microbes multiply in the body, this replication results in the increase of antigenic amount needed for stimulation of the immune system. Also, this antigen remains for a long time in the body, continuous presence of antigen continuously keeps the immune system active for longer time resulting in long term immunity.

Inactivated Killed Vaccines: Sometimes a non-pathogenic version of the microbe is not available, and to make a vaccine in this situation, the organism is killed with formalin or phenol and then used as a killed inactivated vaccine. Because organisms in such a vaccine do not replicate in the body, frequent boosters are required to achieve a good level of immunity (antibodies). Also, remember that this is a killed vaccine, so classical cellular immunity is not created with killed vaccines.

Toxoids: Some diseases are caused by bacterial toxins, and not by bacteria. In such cases, toxin is modified (inactivated) and used as a vaccine. Such a vaccine is known as a toxoid. Tetanus toxoid is a good example.

Conjugated Vaccines: Sometimes special parts of organisms are required for generating protective immunity. However, those substances are poor antigens if used alone. In order to enhance immunity, such poor antigens are attached to proteins, and such vaccines are called conjugated vaccines. Haemophilus influenzae type b is a good example.

DNA vaccines: We now know that sometimes only certain proteins are required for generating a good immune response. In such cases, the genes responsible for those proteins can act as vaccines. DNA coding for such a protein is injected into cells, and that DNA gets transcribed by the cells of the body into the required protein and that protein induces the antibody production (because that protein is foreign to the body). West Nile Virus is a good example of DNA vaccine

Subunit vaccines: Also called recombinant vaccines, such vaccines are composed of a portion of viral or bacterial proteins that can induce immunity. In other words, this is another version of DNA vaccines. In subunit vaccines, antigenic epitopes are included in the genome which is inserted into an expression vector that makes those subunit proteins. These subunit proteins are collected and used as vaccines to stimulate the immune system. Again, these act like a killed vaccine. There is another mechanism being explored in these days for recombinant vaccines. A subunit vaccine for Foot and Mouth disease virus is available.

Such genes are being incorporated in plants now that are acting like an expression vector. Animals can eat those plants and subunit vaccines can induce immune response through M cells of the intestine. Success is limited though so far.

30. Describe features of fungi 5 marks

These are simple eukaryotic cells. Some are unicellular, others are multicellular (thallus); however, they lack tissues such as roots, stem and leaves typically seen in plants. Algae absorb nutrients from water through their surfaces and are mostly photoautotrophs; however, a few are chemoheterotrophs. They are responsible for 80% atmospheric O2 on the face of the earth. Microscopic exam is needed to identify unicellular and filamentous algae. However, multicellular algae that are commonly known as seaweeds are macroscopic in nature and can be identified morphologically without the help of a microscope. Four groups of such algae include blue-green algae, green algae, brown algae and red algae. These algae are located in the sea at various locations and absorb light of various wavelengths, hence red algae are located r from the surface and can use blue light from the sun as blue light is of shorter wavelength and can penetrate deep in the sea. Please see the accompanying figure (above) for a better understanding of how wavelength relates with various algae. Also remember, blue-green algae need magnification in order to be correctly identified, although they are not microscopic.

31. Treatment of hypersensitivity ii 3 marks

This type of hypersensitivity can be diagnosed by history and symptoms. Glomerulonephritis and cellulitis is very common. Deposition of immune complexes in the kidneys can be diagnosed by presence of IgG in kidney cells by using anti-IgG antibodies.

32. What are antibiotics 3 marks

antibiotics are used to kill organisms in disease states, some antibiotics are not very effective for this purpose; however, these antibiotics could be used in food to prevent food spoilage. Nisin and natamycin prevent spoilage of cheese

33. Define virion and viroid 2 marks

A **virion** is an infectious particle composed of nucleic acid that is surrounded by a protein coat (capsid). Viruses are classified based on their nucleic acid and protein coat. Viruses have an extracellular and an intracellular state. Outside of a cell, in the extracellular state, a virus is called a virion.

VIROIDS: They are composed only of circular single stranded (ss) RNA.

34. Write entry point of microorganism 2 marks

Mucus membranes: These include linings of digestive tract, respiratory tract, urogenital system, and conjunctiva

Skin: Surface area wise, skin is the largest organ of the body. Unbroken skin is impermeable to organisms

Parenteral Route: Injury to the skin or mucous membrane leads to easy access for the organisms in the body. Any cut received on the skin for example becomes a portal of entry

35. Role of thin clonal selection 2 marks

According to this theory, B cells are generated in the bone marrow randomly without seeing the antigen. When such a B cell encounters its specific antigen in the body, it starts proliferating into a clone which ultimately becomes plasma cells and start secreting antibodies. A beautiful example to understand clonal selection theory is like going to a shoe store and select a shoe size that fits you very well. Note, that the shoe that you selected was made randomly without any regard to your foot size.

35.Antigen presenting cell 3 marks

These cells are able to present antigens to B cells for making antibodies by B cells.

There are three cells that fall into this category: Dendritic cells, macrophages, and B cells

36.How antibodies provide defense against microbes 5 marks

37.Reverse transcription 2 marks

A reverse transcriptase is an enzyme used to generate complementary DNA from an RNA template, a process termed reverse transcription

38.Bactericidal and bacteriostatic drug 2 marks

Bactericidal: Those antibiotics that kill and lyse the cells are called bactericidal.

Bacteriostatic: Such antibiotics stop the growth of the organisms, they do not kill or lyse the cells. Cells of the immune system then clear them from the body

39. Define plasmogamy karyogamy and meiosis 3 marks

1.Plasmogamy: A haploid nucleus of a donor cell (+) penetrates the cytoplasm of a recipient cell (-). Both cells are haploid (one set of chromosomes)

2. **Karyogamy** The (+) and (-) nuclei fuse to form a diploid zygote nucleus.

3. **Meiosis**: The diploid nucleus gives rise to haploid nuclei (sexual spores).

40. Mechanism of hypersensitivity iv 5 marks

It is called as delayed type because its bad effect becomes visible 2-3 days after exposure to the antigen. This is the only hypersensitivity that involves cells rather than antibodies. Robert Koch discovered this when he injected tuberculin in patients. Contact dermatitis is the most common example of this kind of hypersensitivity. This is basically an interaction between an antigen, T cells and macrophages. Transplant rejection is another good example of delayed type IV hypersensitivity. Reaction to hair dyes is another example. Poison Ivy is another example in which contact with this plant initiate type IV hypersensitivity response. Chemicals from poison ivy interact with skin proteins and change them into foreign-looking protein molecules. Body's immune system mounts an immune response against these proteins resulting in a damaging type of intense inflammatory response

41. Cytokines 2 marks

Cytokines are released as a result of defense cell stimulation by PAMPs and TLRs interactions. Cytokines recruits more immune cells and also stimulate adaptive immune response

42. Note on infectious disease transmission with vector vehicle and contacts?

Direct Transmission: person to person physical contact is a direct contact. Influenza spread is a typical example.

Vehicle Transmission: Transmission of disease agents by a medium, such as water, food, or air. Other media include blood and other body fluids, drugs, and intravenous fluids.

A vector is a vehicle (animate or inanimate) that carries a pathogen from one host to another. o Arthropods are vectors for many diseases. There are two ways these arthropods can transmit organisms to susceptible hosts:

Mechanical Transmission: In this type of transmission, organisms stick to the body parts of insects and get transmitted passively from one location to another.

Biological Transmission: Microbes multiply in the insect and is transmitted actively by insect bites from one individual to another.

43. Incidence and Prevalence 2 marks

Incidence: # of new cases in a given time period in a population

Prevalence: New + existing cases

44. PRIMARY AND SECONDARY IMMUNE RESPONSE 5 marks

When a person is exposed to an antigen (a microbe for example), the antigen is presented to B cells which start making antibodies. Initially, there are no detectable antibodies in the serum. However, after 3-5 days, IgM antibodies are secreted and then 7-10 days after, the same B cell starts making IgG. The level of IgG rises to a level and then it starts declining. This is a typical response of an antigen exposure and is called a primary immune response. It is interesting to note that a second exposure with the same antigen results in a heightened response in terms of very high levels of antibodies that are achieved in a very short time (within 2-7 days). This is called a secondary immune response. It is also called a memory or anamnestic response because memory cells are involved that ensure a quick response to the second exposure by the same antigen (to which the immune system has already been exposed). The accompanying figure illustrates this concept (next page). Blue line in the figure depicts a primary and a secondary response.

Adaptive Immune Response can be acquired naturally or artificially. Each of these categories can further be divided either into active or passive type.

Naturally Acquired Active Immunity: This type of adaptive immunity develops in a person who gets infected with a microbe but then recovers from that illness making antibodies.

Naturally Acquired Passive Immunity: A typical example of this type of immunity is transfer of antibodies from colostrum or milk of a mother to the offspring.

Artificially Acquired Active Immunity: When a person receives a vaccine against a microbe and the body makes antibodies in response to that vaccine, it is called artificial acquired active immunity because body makes these antibodies actively by itself.

Artificially Acquired Passive Immunity: When premade antibodies are given to a person in order to protect a person from an illness or a snake bite, this constitutes an artificially acquired passive immunity type. The following figure illustrates these four types.

45. Vegetative Hyphae and Aerial hyphae 2 marks

Vegetative Hyphae: Hyphae that are embedded in the medium and are used to obtain food are called vegetative hyphae.

Aerial or Reproductive Hyphae: The portion of hyphae that is concerned with reproduction. This portion is projected into the air. They bear reproductive spores.

46. Signature sequence and primer

signature sequences

There are many tests that involve nucleic acid sequence detection for the presence of microbes in a clinical sample. Please remember, as antibodies are specific to the organisms, DNA or RNA sequences must be uniquely specific for the microbe that we are going to detect. Such sequences that are specific to an organism are called **signature sequences**

Primer

A primer is a fragment of DNA which is specific to the signature sequence

Feature of yeast cells

47. Active and passive transport

active Movement

Movement against the concentration gradient

From low concentration to high concentration of a substance

Requires energy (ATP)

Substances that are actively transported into the cells undergo either no change or they are chemically modified to keep them there in the cell. Remember, active transport brings substances inside the cell against the gradient; hence molecules brought in can go out. That is why these molecules may be modified to keep them inside the cells. In other words, this modification avoids passive diffusion of those molecules from the cells outwards.

Passive Movement

Movement of substances with the concentration gradient. This means that substances will move from higher concentration of a substance to lower concentration. Gradient provides the force for movements of molecules and this happens with no energy expenditures.

What are the 2 benefits algae get from oceans 2 marks

Algae are important for aquatic life. They are at the bottom of the food chain because they fix CO2 into organic molecules (Carbs) which could be used by other organisms.

They produce 80% of molecular oxygen on earth.

Algal blooms which result from fluctuation of temperature sometimes may cause problems for nature but that happens occasionally.

Algal blooms tell us about the status of pollution in the environment. Planktonic organisms (dead and decaying) are responsible for oil production. Petroleum and natural gas are the residues (hydrocarbons) of diatoms and other planktonic organisms.

Define lichens 2 marks

Lichens are a combination of green alga (or a cyanobacterium) and a fungus. Green alga or cyanobacterium is also called a phycobiont and the fungus is called a mycobiont in this relationship. Both partners have a symbiotic relationship with each other

Hypersensitivity type 1 3 marks

It involves IgE and it occurs within 30 minutes of exposure to the allergen. Antigens called allergens preferentially induce IgE production instead of IgG against the antigens. These IgEs get bound to mast cells in the tissues and a second exposure to the same allergen activates mast cells to secrete histamine and other mediators of inflammation. Shock from drug reactions, venoms, and common allergens causing asthma are some examples of type I hypersensitivity.

Hypersensitivity can be diagnosed by demonstration of IgE by intradermal injections of allergens one by one. If specific IgE are present in the serum, a reaction will be visible on the skin locally in the form of a wheel of inflammation

Transmission of disease 5 marks

Direct Transmission: person to person physical contact is a direct contact. Influenza spread is a typical example.

Indirect Transmission: A nonliving object is involved. Objects in the use of a patient such as handkerchief, utensils, pillow and bedding are called fomites. These can transmit organisms to susceptible individuals.

Droplet Transmission: Mucus droplets are created when you sneeze for example. These droplets carry

organisms that may infect another individual who comes in contact with the droplets. One sneeze may produce 20,000 droplets. Influenza can be spread by this route.

Vehicle Transmission: Transmission of disease agents by a medium, such as water, food, or air. Other media include blood and other body fluids, drugs, and intravenous fluids.

Hypersensitivity type3 5 marks

This hypersensitivity involves soluble antigen circulating in the serum and antibodies (IgG or sometimes IgM). Complement gets activated when these complexes get deposited in capillary beds as we see in the kidneys for example. A typical example will be the use of antivenom (antibodies) that was prepared in horses but given to a human to save the life. What happens is that horse antibodies will be recognized by human body as a foreign protein and human immune system will generate antibodies against this. This type of hypersensitivity can be diagnosed by history and symptoms. Glomerulonephritis and cellulitis is very common. Deposition of immune complexes in the kidneys can be diagnosed by presence of IgG in kidney cells by using anti-IgG antibodies.

BEST OF LUCK